

Fetal Intracranial Abnormality Detection from Ultrasound Images Using Attention U-Net Segmentation and DenseNet-Based Classification

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Abstract

Fetal intracranial abnormalities are such serious conditions that pose an impact on the structural development of the fetal brain and demand prompt and correct diagnosis to prevent inefficient clinical intervention. Nonetheless, ultrasound imaging that is commonly used in fetal monitoring, has poor contrast, speckle noise, and complicated anatomical structures, which complicates the process of manual diagnosis and time-consuming. We present in this paper a complete deep learning framework that can be used to detect fetal intracranial abnormalities based on ultrasound images. First, Otsu thresholding is used to create pseudo ground truth masks which make it possible. Segmentation training which is weakly supervised. This is followed by an Attention U-Net model that is used to precisely segment abnormal brain areas through highlighting of the features of interest and muting out the background noise. The broken output is further fed on through region of interest (ROI) removal to improve the performance of classification. After that, several deep learning classifiers, such as They are a baseline Convolutional Neural Network (CNN), EfficientNetB0, and DenseNet121. Experimental findings show that DenseNet121 is the most accurate in terms of classification of about 85 percentage better than the other two. other models, which the segmentation model has a Dice score of 0.83 and an IoU score of 0.71. The proposed system is built into an easy-to-use dashboard to be viewed and predictively. In general, the framework will give an efficient structure. and valuable automated fetal abnormality detector, whose possible use would be in clinical decision support system.

Index Terms

Fetal Abnormality Detection, ultrasound, Attention U-Net, Deep Learning, DenseNet121, Image Segmentation, Medical Imaging.

I. INTRODUCTION

Fetal intracranial abnormalities refer to severe developmental conditions, which impact the form and activity of the fetal brain in the course of pregnancy. These abnormalities should be identified early to ensure that medical intervention, treatment planning, and risk assessment occur in good time. The most common modality that is applied in prenatal screening is ultrasound imaging because of the safety, availability, and affordability of the technology. Nevertheless, low contrast, speckle noise, and other complex anatomical variations can greatly harm ultrasound images and thus a correct diagnosis can only be achieved by experienced radiologists with the use of deep learning techniques within the recent years, especially when it comes to the processing of medical images like segmentation and classification. Convolutional Neural Networks (CNNs) have been popular in automated feature extraction and classification, and such architectures as U-Net have shown good results in biomedical image segmentation. Moreover, more sophisticated models like Attention U-Net are better at segmenting objects by focusing on the important regions and trying to suppress the background data that is not relevant. On the same note, the transfer learning frameworks like EfficientNet and DenseNet have been effective in identifying sophisticated patterns with small medical datasets. Nonetheless, there are still a number of challenges in detecting fetal abnormalities. Among the significant limitations, the absence of annotated segmentation data, which is a mandatory requirement of supervised learning, can be cited. Also, most of the current methods do not combine segmentation and classification together but still do it separately in contrast to each other. Additionally, the differences in fetal structure and imbalance of classes amid various abnormalities make the detection process even more complex to handle. In a bid to overcome these problems, this paper introduces an end-to-end deep learning architecture of the detection of intracranial abnormality in fetuses using the ultrasound images. In the proposed method, Otsu thresholding is used to produce pseudo ground truth masks that allows weakly supervised training of segmentation. Abnormal regions are further segmented with the help of an Attention U-Net model. The segmented output is also optimized by extracting region of interest (ROI) which improves the quality of features to classify. The key findings of the work include: (i) combination of Otsu-based pseudo labeling with Attention U-Net to achieve high results of segmentation, (ii) the classification based on ROI, which proves to be more accurate, and (iii) a general analysis of deep learning models, which indicates that DenseNet121

is more effective. The proposed system also will consist of a visualization dashboard to be used in real-time prediction and analysis, which makes it applicable to a practical clinical setting.

II. PROBLEM DESCRIPTION

The fetal intracranial abnormalities are emergencies that may severely impact the development of the brain and may cause severe neurological disorders in case they are not identified early. The use of ultrasound in prenatal screening is a common practice but proper identification of the abnormality based on ultrasound pictures is still a daunting task because of various limitations attached to it. Such images are usually of low contrast, speckle noise, and blurred boundaries making it hard to clearly tell between the abnormal regions and normal brain structures. Also, diagnosis is time-consuming and subject to subjective variability since the interpretation of ultrasound images strongly relies on the skills of radiologists, it has been noted that there is scarcity of annotated datasets particularly in the segmentation tasks which demand accurate pixel-level labels. Such a lack of labeled data limits the usefulness of the supervised learning methods. Moreover, the current practices usually revolve around segmentation or classification as it is, instead of combining both processes in a common framework. Consequently the general accuracy and reliability of fetal abnormality detection systems are a concern and thus there is a need to come up with an automated, efficient and reliable system that can be able to segment and classify fetal intracranial abnormalities based on ultrasound images without interference by questions like noise, poor image quality and limited labelled data.

III. LITERATURE REVIEW

The last improvement in deep learning has enhanced the artificial analysis of fetal ultrasound images to detect intracranial abnormality in a significant way. Wu et al. [1] suggested a CNN-based system that automatically predicted the intracranial abnormalities of the fetus, and it demonstrated a high diagnostic performance. With the use of deep learning methods, Namburete et al. [2] were able to isolate fetal brain biometrics, including ventricular width, which allows an object to quantitatively evaluate abnormalities. According to Zeng et al., a deep learning model of fetal brain abnormalities was presented, with high-quality screening results . [3]. Ravindra et al. [4] created an automated fetus central nervous system anomaly system, eliminating the use of experts. Moreover, Li et al. [5] and Chen et al. [6] were interested in determining conventional ultrasound planes of the fetal brain by means of deep convolutional networks, which constitutes one of the most significant preprocessing stages to make the correct diagnosis. Looney et al. (7) in the segmentation field introduced an auto- CNN model of fetal brain structure segmentation, and Ding et al. (8) introduced a model of cerebellum segmentation that identifies abnormalities in the posterior fossa. Comprehensive surveys that have been made in the literature have also widely proved the effectiveness of deep learning in medical imaging (see [9], [10]. Moreover, the ISUOG guidelines define standardized protocols of acquiring fetal ultrasound which is important in a consistent clinical diagnosis [11]. Irrespective of these improvements, the current practices continue to experience major setbacks such as class imbalance, absence of annotated datasets, restricted category of abnormalities, and real-time clinical implementation challenges. As such, a complete and robust structure is required to counter such limitations and enhance the segmentation as well as classification performance.

IV. OBJECTIVES OF THE STUDY

The primary goal of this study is to come up with a high-precision deep learning-based framework that can be used to identify the existence of intracranial pathologies in the fetus using ultrasound images. The paper will be trying to improve the quality of the images by using the preprocessing methods to remove noise, enhance contrast and standardise the inputs to optimise the model. It also pays attention to the comparison of various deep learning architectures such as CNNs, transfer learning models, and hybrid models and finding the most effective model in detecting abnormalities. In order to overcome the limitations of the dataset, data augmentation and class balancing are used to address the variation in the fetal position, gestational age, and image quality. To analyze the quality of medical interpretation, the proposed system will be analyzed in terms of clinically relevant performance metrics, including accuracy, sensitivity, specificity, and precision. The work is also intended to decrease the operator dependency by allowing reliable analysis of variable image acquisition conditions, integrate explainability methods to visualize model decisions and facilitate clinical interpretation, and create a workflow to be scaled to real-time and clinically deployable systems to detect early in clinically practice-relevant screening by routine prenatal ultrasound.

V. PROPOSED METHODOLOGY

A. SYSTEM OVERVIEW

This paper suggests a complete deep learning approach to the automated detection of intracranial abnormalities in fetuses through ultrasound images. The general layout incorporates the preprocessing, pseudo mask generation, segmentation, region of interest (ROI) extraction and classification in a single pipeline. The problems that the proposed system aims at solving include poor quality of the image, image noise and that the annotated segmentation data of fetal ultrasound imaging is not available, to begin with, the proposed system aims to preprocess the input images by first resizing them to a standard resolution and secondly normalizing the pixel intensities of the images, so that there is uniformity across the dataset. Otsu thresholding has

been used to create pseudo ground truth masks as manually annotated segmentation masks are not easily available. This loosely controlled method allows proper training of the segmentation model without the need of the massive manual annotation. This is followed by an Attention U-Net model which segmentation of the fetal brain areas is carried out. The attention mechanism enables the model to attend to the relevant anatomical structures whilst blocking background noise thus enhancing segmentation accuracy. The result of the segmentation phase is binary mask which emphasizes the abnormal areas of the fetal brain. After segmentation, the region of interest (ROI) is obtained out of the outcome of the predicted mask. This is done to eliminate the irrelevant background data and only the significant features related to the abnormality are retained increasing the usefulness of the classification models. In the classification phase, several deep learning models are trained and tested, such as a baseline Convolutional Neural Network (CNN), EfficientNetB0 and DenseNet121. These models are trained using the extracted ROI images in order to model the fetal abnormalities based on already set categories. Of them, DenseNet121 is the most successful in its performance because of its dense connectivity and reusing features. Lastly, the visualization of the predicted results is performed in the form of an interactive dashboard that shows the original ultrasound image, generated masks, segmented output, predicted class, and confidence scores. The entire system offers an automated and wholesome approach to the detection of fetal intracranial abnormality; that is, segmentation and classification are both incorporated within a single framework.

B. PREPROCESSING

The preprocessing process plays a significant role in the analysis of the medical images; especially when it comes to the ultrasound images that are usually characterized by noise, low contrast and changes in intensity. Preprocessing In the present study, the preprocessing is done to optimize the quality of images, standardize the data set, and optimize the performance of the further segmentation and classification models by resizing all input ultrasound images to a specified resolution of 256x256 pixels to provide consistency and alignment to the dataset and to the requirements of the input to the deep learning models. This resizing process is also necessary to simplify the computation and maintain the most important anatomical features. Normalization of pixel intensities is also performed after resizing to bring the image value between the range of [0,1]. This is done by dividing each pixel value with the largest intensity value as provided, in addition data augmentation is used to make the training data more diverse and enhance the ability of the models to generalize. These methods are rotation, horizontal flipping, zooming, and translation which mimics changes in fetal position and imaging parameters. To a large extent, data augmentation will play a significant role in this research because there is a scarcity of labeled medical images, and it will help to standardize the images used in the study, make them noise-tolerant, and make them valuable to efficient feature extraction and the overall performance of the proposed framework, serving segmentation, and classification functions.

C. Otsu Thresholding for Pseudo Mask Generation

With medical image segmentation, ground truth masks that have been accurately annotated are required in supervised learning. In the case of fetal ultrasound imaging, however, pixel-level annotations can hardly be obtained because it requires expert knowledge and manual labeling can be time-consuming. In a bid to overcome this, a weakly supervised position is taken in this study by using Otsu thresholding to produce pseudo ground truth masks.

Otsu thresholding is an automatic pixel partitioning algorithm that will compute a desirable threshold value making a maximum of the variation between foreground and background pixel values. It presumes that the image has two different types of pixels and calculates a threshold that can effectively divide these types of pixels. The algorithm reduces the intra-class variance and maximizes the inter-class variance to create an effective binary segmentation.

Mathematically, the optimal threshold t^* is selected such that:

$$t^* = \arg \max_t \sigma_b^2(t) \quad (1)$$

where $\sigma_b^2(t)$ represents the between-class variance, defined as:

$$\sigma_b^2(t) = \omega_0(t)\omega_1(t) [\mu_0(t) - \mu_1(t)]^2 \quad (2)$$

Here, ω_0 and ω_1 denote the probabilities of the background and foreground classes, respectively, while μ_0 and μ_1 represent their corresponding mean intensities.

In this paper, we use Otsu thresholding on grayscale ultrasound images to produce the binary mask that will approximate the regions of interest. These masks are pseudo ground truth labels which are used to train the Attention U-Net segmentation model. Even though the masks generated may not be very precise, they give enough structural information that can direct the learning process.

Otsu-based pseudo labeling eliminates the reliance on manual labeling and makes the segmentation model training efficient. This method is especially useful in medical imaging cases where labeled data is limited, thus it is able to develop scalable and cost-effective models. —human—;section U-Net Segmentation.

Attention U-Net architecture is used to segment the image to be able to identify the abnormal regions in fetal ultrasound images properly. Attention U-Net is an extension of the original U-Net architecture that adds attention to the architecture to enhance localization of the relevant features and suppression of irrelevant background data.

The U-Net model is divided into two primary components, including an encoder (contracting path) and a decoder (expanding path). The encoder takes into account the information of the context through successive convolution and pooling operations and the decoder recreates spatial resolution with upsampling and skip connection. In normal U-Net, however, the encoder directly transmits all of the features to the decoder, which could introduce irrelevant information.

In order to overcome this shortcoming, attention gates are included in the skip connections. These attention gates enable the model to concentrate on significant areas by giving priority on significant features and inhibiting the less critical areas. The mechanism enhances the performance of segmentation especially in complicated medical images like fetal ultrasound scans.

The attention coefficient α is computed as:

$$\alpha = \sigma (W^T (\psi(x_l) + \phi(g)) + b) \quad (3)$$

where x_l represents the feature map from the encoder, g denotes the gating signal from the decoder, ψ and ϕ are linear transformations, W and b are learnable parameters, and σ is the sigmoid activation function.

The output of the attention gate is obtained by:

$$\hat{x}_l = \alpha \cdot x_l \quad (4)$$

where $\hat{x}_l$ is the refined feature map that emphasizes relevant regions.

In the present research, Attention U-Net is trained on pseudo masks created in Otsu thresholding. By emphasizing significant structural parts of the fetal brain and minimizing noise and background artifacts, the model learns to divide abnormal parts of the fetal brain.

The Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) are used to assess the performance of segmentation model. The Dice score is defined as:

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|} \quad (5)$$

and the IoU score is given by:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \quad (6)$$

where A represents the ground truth mask and B denotes the predicted mask.

Attention U-Net model proves to be a highly performing model in the segmentation of fetal intracranial abnormalities with high scores of Dice and IoU thus giving precise abnormal area localization to be further classified.

D. Region of Interest (ROI) Extraction

Once the segmented output of the Attention U-Net model is obtained, the region of interest (ROI) is isolated to target the areas of interest of the fetal brain abnormalities. The extraction of ROI is important in enhancing the performance of the classification by eliminating the irrelevant background facts and retaining the relevant ones that are related to abnormalities.

The abnormal region is isolated by the predicted segmentation mask of the original ultrasound image. This is done by the element-wise multiplication of the input image and the binary mask which is defined as:

$$I_{ROI} = I \cdot M \quad (7)$$

where I represents the input image, M denotes the predicted binary mask, and I_{ROI} is the extracted region of interest.

In order to generalize the ROI further, bounding box cropping is used on the segmented area to remove unnecessary background pixel. The resulting images of ROI are then down-sampled to fit the dimensions of the input needed by the classification models.

The ROI extraction step amplifies the discriminative characteristics of the abnormal areas, which makes the classification stage, which follows the extraction, more accurate and robust. This will ensure that the classification models only consider the clinically relevant information to minimize the effects of noise and irrelevant structures that exist in the ultrasound images.

The processed images are then classified to detect various forms of fetal intracranial abnormalities after the extraction of ROI. Three deep learning models (Convolutional Neural Network (CNN) as a baseline, EfficientNetB0, and DenseNet121) are applied and compared in this study.

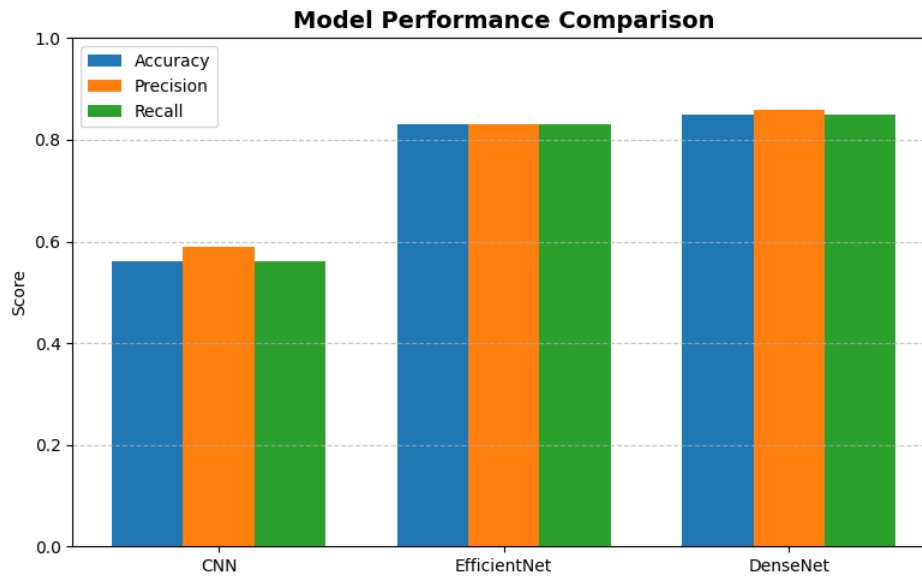


Fig. 1. comparison graph

1) *CNN (Baseline Model)*: A standard Convolutional Neural Network is considered a baseline model of classification. The CNN is made up of several convolutional layers and pooling layers which are used to extract the hierarchical features of the images used. These characteristics are then classified by taking them through fully connected layers. Despite the simplicity and effectiveness of CNN approach, the data performance of this method is weak in the capture of complicated patterns in medical images. .

2) *EfficientNetB0*: EfficientNetB0 is a modern deep learning network, which employs compound scaling to trade depth, width and resolution of the network. In this paper, an EfficientNetB0 model, which has been pre-trained, is fine-tuned with the help of transfer learning. This enables the model to use learned characteristics of big datasets to enhance the accuracy of classification and to traverse shorter periods in training

3) *DenseNet121*: DenseNet121 proposes the use of dense layers with the connection between layers, meaning that each layer is fed with the outputs of all the previous layers. Such design encourages reuse of features and gradient flow, which causes better performance in learning. DenseNet121 is one model that is especially useful in medical image classification problems because it is able to capture complex and subtle patterns in the data.

All the models are trained on ROI images and the accuracy, precision, recall and F1-score are the performance metrics that are used to assess the performance of the models. Of all the models, the DenseNet121 model is the top one, as it exhibits the highest classification accuracy.

E. Model Comparison

The three models of classification are CNN, EfficientNetB0, and DenseNet121 and a comparative analysis is made to determine the effectiveness of the proposed framework. The models are evaluated by their accuracy, precision, recall and F1-score.

The experimental findings suggest that the baseline CNN has moderate performance because of the low feature extraction ability. EfficientNetB0 is an enhanced version that makes use of transfer learning and scaling compounds. Nevertheless, DenseNet121 is better than the two models in that it is able to capture complex features in an effective way because of dense connectivity..

The findings show clearly that DenseNet121 is the best performing model when it comes to the performance of the models that have been tested. Consequently it gets chosen as the last model to use in the classification of fetal intracranial abnormality.

TABLE I
PERFORMANCE COMPARISON OF CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN	56%	0.59	0.56	0.54
EfficientNetB0	83%	0.83	0.83	0.83
DenseNet121	85%	0.86	0.85	0.85

VI. EXPERIMENTAL RESULTS AND DISCUSSION

This section displays the performance analysis of the suggested framework of fetal intracranial abnormality detection. In the system, there are two phases of evaluation: segmentation and classification.

A. Segmentation Results

The Attention U-Net model is measured based on the segmentation performance in the form of Dice Similarity Coefficient (DSC) and Intersection over Union (IoU), and other measures including precision, recall, and accuracy.

TABLE II
SEGMENTATION PERFORMANCE OF ATTENTION U-NET

Model	Dice	IoU	Precision	Recall	Accuracy
Attention U-Net	0.83	0.71	0.85	0.82	0.88

The findings show that the Attention U-Net model is effective in the segmentation of abnormal regions in fetal ultrasound images with a high Dice score of 0.83 and an IoU score of 0.71.

B. Classification Results

Three models are used to assess the classification performance including CNN, EfficientNetB0, and DenseNet121. Accuracy, precision, recall and F1-score are used to evaluate the models.

The findings indicate that DenseNet121 has the best classification accuracy of 85 percent as compared to CNN and EfficientNetB0.

C. Confusion Matrix Analysis

To further analyze classification performance, the confusion matrix of the best-performing model (DenseNet121) is presented in Fig. 2.

The confusion table shows that a majority of the classes are well classified as depicted by high values on the diagonal. Minor misclassifications are noted between the similar categories of abnormalities especially the different types of ventriculomegaly. This is primarily because there is similarity in visual features and class imbalance in the dataset.

D. Discussion

The results of the experiments show that the suggested framework can be successfully used to integrate segmentation and classification to detect fetal abnormalities. Otsu thresholding allows generating pseudo mask, and Attention U-Net enhances the accuracy of segmentation by targeting important areas. The extraction of ROI also improves the performance of classification by minimizing the noise in the background.

DenseNet121 is better than CNN and EfficientNetB0 because it has dense connectivity and efficient feature-reuse, similarity, and it can easily learn more intricate patterns in ultrasound images. In general, the suggested system is a good and productive tool to have in place to detect the occurrence of fetal intracranial abnormality in an automated manner.

VII. FUTURE WORK

Despite the successful outcomes of the proposed framework, there are some perspectives to improve it in the future. To begin with, bigger and more heterogeneous datasets could be used to increase the generalization property of the model and decrease the problem of class imbalance. Second, it is possible to consider more sophisticated deep learning frameworks including Vision Transformers and hybrid models that can improve the classification performance further.

Third, the accuracy of the segmentation can be improved with the use of manually annotated ground truth masks as opposed to pseudo masks created with the help of thresholding methods. Also, the real-time implementation of the system in the clinical settings can be created to help the radiologists in diagnosis.

TABLE III
CLASSIFICATION PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score	Support
CNN	0.56	0.59	0.56	0.54	157
EfficientNetB0	0.83	0.83	0.83	0.83	157
DenseNet121	0.85	0.86	0.85	0.85	157

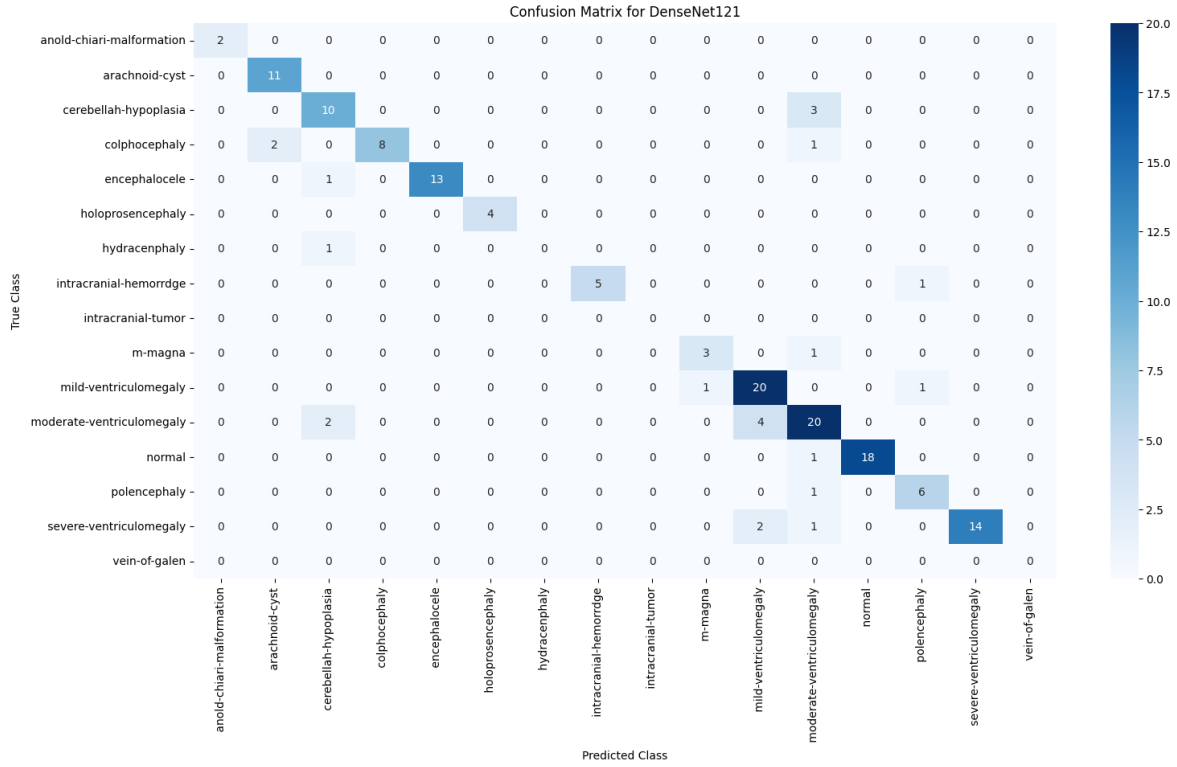


Fig. 2. Confusion Matrix of DenseNet121 Model

Lastly, multi-modal medical data integration, including MRI and clinical reports, are additional ways that can enhance the strength and credibility of the system and can be more applicable to the real-life practices of healthcare.

VIII. CONCLUSION

The proposed end-to-end deep learning model (in the given study) will be applied to the automated detection of fetal intracranial abnormalities given ultrasound images. The proposed system incorporates Otsu-based pseudo mask generation, Attention U-Net segmentation, ROI extraction, and deep learning-based classification and the Attention U-Net model has shown good results, with a Dice score of 0.83 and an IoU score of 0.71, which implies that the abnormal regions are well localized. Three models were considered in order to classify them, such as CNN, EfficientNetB0, and DenseNet121. DenseNet121 with the highest accuracy of 85 percent was the most successful among them because of the dense connectivity and the effective reuse of features. The experimental findings support the hypothesis that the combination of segmentation and classification is much more effective in the overall detection. The system that has been developed lowers the reliance on manual diagnosis and offers an effective and dependable instrument of early identification of fetal brain deformities. Moreover, the system is made easier to use by the inclusion of a visualization dashboard to make real-time clinical analysis easier.

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